Diagrams

Calepin can run text-to-diagram engines from the same chunk system used for Python, R, and other computational code. Diagram chunks keep the source for figures in the Typst document, so prose, references, and diagram definitions stay together.

Diagram engines are stateless external tools. They convert source text to SVG, and *Calepin* renders the SVG as a figure. Give a diagram chunk a `label` and `fig-caption` when it should be numbered and cross-referenced.

Mermaid

Mermaid is a text-based diagramming tool; learn more at mermaid.js.org.

```
%%{init: {"htmlLabels": false}}%%
flowchart LR
  Source["Typst document"] --> Query["Calepin query"]
  Query --> Execute["Run chunks"]
  Execute --> Render["Typst render"]
```

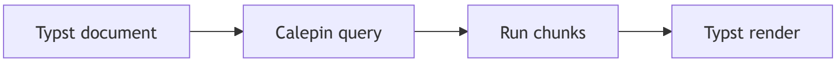

Figure 1: Mermaid flowchart

Graphviz DOT

Graphviz DOT is a graph description language used by Graphviz; learn more at graphviz.org.

```
digraph {
  rankdir=LR
  Draft -> Review -> Publish
  Review -> Draft [label="revise"]
}
```

Figure 2: Graphviz state graph

TikZ

TikZ is a LaTeX package for creating vector graphics; learn more at tikz.dev.

```
\begin{tikzpicture}
  \draw[thick, blue] (0,0) -- (2,1) -- (4,0);
  \fill[red] (2,1) circle (2pt);
\end{tikzpicture}
```

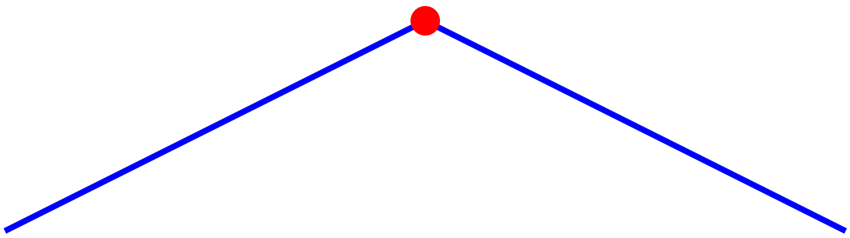

Figure 3: TikZ path

D2

D2 is a text-to-diagram language; learn more at d2lang.com.

```
direction: right

client -> api: request
api -> worker: job
worker -> database: write
```

Figure 4: D2 service sketch